The first, using the `cp` command, takes as input the name of the file to copy and the name of the file to copy it to. The second, using the `cat` command, instead takes as input two *file descriptors*; one opened for reading and the other opened for writing. It then simply reads the data from the first file descriptor and writes it to the second.

> **DEFINITION** A ***file descriptor*** is an abstract handle that represents an open file along with a set of permissions on that file. File descriptors are a type of capability.

If you think about the permissions that each of these commands needs, the `cp` command needs to be able to open any file that you can name for both reading and writing. To allow this, *UNIX runs the* `cp` *command with the same permissions as your own user account*, so it can do anything you can do, including deleting all your files and emailing your private photos to a stranger. This violates POLA, because the command is given far more permissions than it needs. The `cat` command, on the other hand, just needs to read from its input and write to its output. It doesn't need any permissions at all (but of course UNIX gives it all your permissions anyway). A file descriptor is an example of a *capability*, because it combines a reference to some resource along with a set of permissions to act on that resource.

Compared with the more dominant identity-based access control techniques discussed in chapter 8, capabilities have several differences: